

SQL-COSMICDB-Git / cosmic_sql_query-git.ipynb  savinshynu adding scripts and notebooks3abfe9c · 6 months ago  History

Preview

Code

Blame

4117 lines (4117 loc) · 245 KB

Raw



In this Jupyter notebook, we will use SQL queries to understand the structure of the COSMIC database hosting the output data products from the COSMIC SETI pipeline. The overall structure of the COSMIC database is shown in this [link](#).

Let's connect to the database first for that. For connecting to SQL database with python, make sure to install *ipython-sql* and *pymysql* libraries.

In [23]:

```
%load_ext sql
```

The sql extension is already loaded. To reload it, use:

```
%reload_ext sql
```

In [24]:

```
#Use the below format to connect to any database  
#Actual account information retracted for security  
# %sql mysql+pymysql://user:password@Host/database
```

Out[24]: 'Connected: cosmic@cosmic_observations'

Now let's run SQL commands to understand the database

In [25]:

```
%sql SHOW TABLES;
```

```
* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations  
22 rows affected.
```

Out[25]: **Tables_in_cosmic_observations**

cosmic_calibration_antenna_result

cosmic_calibration_antenna_result_pre_20230410

cosmic_configuration_antenna

cosmic_configuration_antenna_pre_20230410

cosmic dataset

cosmic_dataset_pre_20230410
cosmic_observation
cosmic_observation_beam
cosmic_observation_beam_pre_20230410
cosmic_observation_calibration
cosmic_observation_calibration_pre_20230410
cosmic_observation_configuration
cosmic_observation_configuration_pre_20230410
cosmic_observation_hit
cosmic_observation_hit_pre_20230410
cosmic_observation_pre_20230410
cosmic_observation_stamp
cosmic_observation_stamp_pre_20230410
cosmic_observation_subband
cosmic_observation_subband_pre_20230410
cosmic_scan
cosmic_scan_pre_20230410

There are 22 tables of which 11 tables are the same from a different time period. For the purpose of this tutorial, we will focus on tables not ending with pre_pre_20230410. Let's look at the first 5 contents from each of the table.

In [26]:

```
%sql SELECT * FROM cosmic_calibration_antenna_result LIMIT 5;
```

```
* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations
5 rows affected.
```

Out[26]:

calibration_id	antenna_name	tuning	coarse_channels_processed	coarse_channe
6103	ea01	BD	416	
6103	ea02	BD	416	
6103	ea03	BD	416	
6103	ea04	BD	416	
6103	ea05	BD	416	

This shows the calibration id of the observation and number of coarse channels processed and flagged per tuning per antenna.

In [27]:

```
%sql SELECT * FROM cosmic_configuration_antenna LIMIT 5;
```

```
* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations
5 rows affected.
```

Out[27]:

configuration_id	name
11142	ea01
11142	ea02
11142	ea03
11142	ea04
11142	ea07

In [28]:

```
# shows the dataset id for the observations == VLA observation id
%sql SELECT * FROM cosmic_dataset LIMIT 5;
```

```
* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations
5 rows affected.
```

Out[28]:

id
10995_1_sb41827221_1_1.60171.9014428588
14A-000.sb43728141.eb43813141.60048.41892384259
14A-000.sb43728848.eb43827735.60055.33266231482
14A-000.sb43813282.eb43832397.60062.30340255787
14A-000.sb43813989.eb43862811.60069.277431215276

In [29]:

```
# More observation of scan_id, time stamps and the recording criteria =
%sql SELECT * FROM cosmic_observation LIMIT 5;
```

```
* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations
5 rows affected.
```

Out[29]:

id	scan_id	configuration_id
1	23A-195.sb43224454.eb43786891.60044.483839548615.11.1	67 20 12:18:3
2	23A-195.sb43224454.eb43786891.60044.483839548615.11.1	68 20 12:19:3
3	23A-195.sb43224454.eb43786891.60044.483839548615.13.1	69 20 12:25:4
4	23A-195.sb43224454.eb43786891.60044.483839548615.17.1	70 20 12:43:0

In [30]:

Listing the cosmic beam observation, coordinates, what the source was
Most of these observations are roughly 8 seconds
%sql SELECT * FROM cosmic_observation_beam LIMIT 5;

* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations
5 rows affected.

Out[30]:

	id	observation_id	ra_radians	dec_radians	source	
	1	1195	1.2316924929618835	0.7853981633974483	PHASE_CENTER	14
	2	1195	1.2317071557044983	0.7853981633974483	PHASE_CENTER	14
	3	1196	1.037502110004425	0.7853981633974483	PHASE_CENTER	14
	4	1196	1.0375168025493622	0.7853981633974483	PHASE_CENTER	14
	5	1197	1.9453375339508057	0.7853981633974483	PHASE_CENTER	14

In [31]:

#Let's look at how many distinct source we observed.
%sql SELECT COUNT(DISTINCT(source)) as Observed_Sources FROM cosmic_obs

* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations
1 rows affected.

Out[31]:

Observed_Sources
269285

So in total database has observations of 269364 distinct sources. Let's see how many of them are actually PHASE Center ones.

In [32]:

%sql SELECT COUNT(source) as Observed_Sources FROM cosmic_observation_b

* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations
1 rows affected.

Out[32]:

Observed_Sources
7210

In [33]:

```
# Looking at the contents of the calibration table
```

```
%sql SELECT * FROM cosmic_observation_calibration LIMIT 5;
```

```
* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations  
5 rows affected.
```

```
Out[33]:
```

id	observation_id	reference_antenna_name	overall_grade	
1	4810	ea27	0.7988652489713474	2/data1/uvh5_
2	4811	ea27	0.8860821601507426	2/data1/uvh5_
3	4837	ea27	0.03386563507387843	2/data1/uvh5_
4	4846	ea27	0.8078979568801078	2/data1/uvh5_
5	4856	ea27	0.03862982513115466	2/data1/uvh5_

This table lists the observation id, reference antenna used for gain calibration, overall grade and the storage location of the correlated uvh5 file used for calibration

```
In [34]: # observation configuration table
```

```
%sql SELECT * FROM cosmic_observation_configuration LIMIT 5;
```

```
* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations  
5 rows affected.
```

```
Out[34]:
```

id	scan_id	start
----	---------	-------

1	23A-	2023-04-10	2023
	195.sb43224454.eb43786891.60044.483839548615.1.1	11:36:47.596100	11:37:13.

2 23A- 2023-04-10 2023
195.sb43224454.eb43786891.60044.483839548615.2.1 11:37:45.400616 11:37:59.

4 23A- 2023-04-10 2023
195.sb43224454.eb43786891.60044.483839548615.2.1 11:40:23.671158 11:40:39.3

5 23A- 2023-04-10 2023-04-10
195.sb43224454.eb43786891.60044.483839548615.2.1 11:40:54.375831 11:41:09.1

This table lists all observation id, times and details cosmic data collection and processing engine during observations.

Now let's look at the hits table >> which has the list of all the interesting technosignature candidates output from the seticore search part the COSMIC pipeline.

```
In [35]: # Hits table
%sql SELECT * FROM cosmic_observation_hit LIMIT 50;

* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations
50 rows affected.
```

Out[35]:

id	beam_id	observation_id	tuning	subband_offset	
1	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
2	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
3	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
4	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
5	12	1202	BD2	480	

5	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
6	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
7	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
8	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
9	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
10	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
11	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
12	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
13	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
14	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
15	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
16	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
17	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
18	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
19	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
20	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
21	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
22	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
23	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
24	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
25	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168

26	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
27	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
28	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
29	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
30	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
31	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
32	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
33	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
34	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
35	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
36	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
37	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
38	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
39	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
40	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
41	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
42	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
43	12	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
44	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
45	11	1202	BD2	480	2/data0/discard/TLIE/TLIE.168
46	11	1202	AC2	512	2/data0/discard/TLIE/TLIE.168

47	12	1202	AC2	512	2/data0/discard/TLIE/TLIE.168
48	11	1202	AC2	512	2/data0/discard/TLIE/TLIE.168
49	12	1202	AC2	512	2/data0/discard/TLIE/TLIE.168
50	12	1202	AC2	512	2/data0/discard/TLIE/TLIE.168

In [36]:

```
# The above table has all the attributes of the hits table
# look at number of hits in the table with frequency greater than 10000
# greater than 500.
%sql SELECT COUNT(id) as Number_hits \
FROM cosmic_observation_hit \
WHERE signal_frequency > 10000.0 AND signal_snr > 500.0;
```

* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations
1 rows affected.

Out[36]: **Number_hits**

2120989

Now look at the Stamp table which has snippets of raw voltage data saved for all the interesting. The stamps also has the metadata of the hit for which the data is being saved.

In [37]:

```
# Stamps table
dat = %sql SELECT * FROM cosmic_observation_stamp LIMIT 5;

# Reading the query results into a database
df = dat.DataFrame()
print(df.head)
```

* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations
5 rows affected.

```
<bound method NDFrame.head of      id  observation_id tuning  subband_offs
et  \
0    1           1202    BD2         480
1    2           1202    BD2         480
2    3           1202    BD2         480
3    4           1202    BD2         480
4    5           1202    BD2         480

                                file_uri  file_local_enumeration \
0  /mnt/cosmic-storage-2/data0/discard/TLIE/TLIE....
0
1  /mnt/cosmic-storage-2/data0/discard/TLIE/TLIE....
1
2  /mnt/cosmic-storage-2/data0/discard/TLIE/TLIE....
```

```

2
3 /mnt/cosmic-storage-2/data0/discard/TLIE/TLIE....
3
4 /mnt/cosmic-storage-2/data0/discard/TLIE/TLIE....
4

source_name  ra_hours  dec_degrees  fchl_mhz  ...  signal_index  \
0      Fake  8.613567      45.0  8671.790089  ...      38089
1      Fake  8.613567      45.0  8687.789310  ...      37987
2      Fake  8.613567      45.0  8699.235546  ...      96474
3      Fake  8.613567      45.0  8698.629406  ...      17027
4      Fake  8.613567      45.0  8699.236454  ...      96593

signal_drift_steps  signal_drift_rate  signal_snr  signal_beam  \
0                -2      -1.847862    31.977867      1
1                -2      -1.847862     9.462310      1
2                 1       0.923931     9.879915      0
3                -1      -0.923931    59.913723      0
4                 1       0.923931    13.985707      1

signal_coarse_channel  signal_num_timesteps  signal_power  \
0                     4                   64  2.443524e+12
1                     20                   64  8.938084e+11
2                     31                   64  9.187028e+11
3                     31                   64  4.046031e+12
4                     31                   64  1.166873e+12

signal_incoherent_power  beam_id
0      2.443524e+12      12
1      8.938084e+11      12
2      7.840300e+11      11
3      3.827953e+12      11
4      1.166873e+12      12

```

[5 rows x 34 columns]>

The Observation Stamp table has 34 attributes associated with each row. Let's look at the contents of the last 2 tables as well.

In [38]: `%sql SELECT * FROM cosmic_observation_subband LIMIT 5;`

```

* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations
5 rows affected.

```

Out[38]:

observation_id	tuning	subband_offset	percentage_recorded	successful_particip
1	AC	512	100.0	
1	AC	544	100.0	
1	AC	576	100.0	
1	AC	608	100.0	
1	AC	640	100.0	

In [39]:

```
%sql SELECT * FROM cosmic_scan LIMIT 5;
```

```
* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations  
5 rows affected.
```

Out[39]:

id

data

10995_1_sb41827221_1_1.60171.9014428588.1.1 10995_1_sb41827221_1_1.60171.901442

10995_1_sb41827221_1_1.60171.9014428588.10.1 10995_1_sb41827221_1_1.60171.901442

10995_1_sb41827221_1_1.60171.9014428588.11.1 10995_1_sb41827221_1_1.60171.901442

10995_1_sb41827221_1_1.60171.9014428588.12.1 10995_1_sb41827221_1_1.60171.901442

```
10995_1_sb41827221_1_1.60171.9014428588.13.1 10995_1_sb41827221_1_1.60171.901442
```

The cosmic scan columns contain the all the important metadata information for the observations.

Now let's look at some observation Stamps which has source names similar to 3C* which are strong 3C sources.

In [40]:

```
%sql DESCRIBE cosmic_observation_stamp;
```

```
* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations
34 rows affected.
```

Out[40]:

	Field	Type	Null	Key	Default	Extra
	id	int	NO	PRI	None	auto_increment
	observation_id	int	NO	MUL	None	
	tuning	varchar(10)	NO		None	
	subband_offset	int	NO		None	
	file_uri	varchar(255)	NO		None	
	file_local_enumeration	int	NO		None	
	source_name	varchar(80)	NO		None	
	ra_hours	double	NO	MUL	None	
	dec_degrees	double	NO	MUL	None	
	fch1_mhz	double	NO		None	
	foff_mhz	double	NO		None	
	tstart	double	NO		None	

tsamp	double	NO	None	
telescope_id	int	NO	None	
num_timesteps	int	NO	None	
num_channels	int	NO	None	
num_polarizations	int	NO	None	
num_antennas	int	NO	None	
coarse_channel	int	NO	None	
fft_size	int	NO	None	
start_channel	int	NO	None	
schan	int	NO	None	
obsid	varchar(100)	NO	None	
signal_frequency	double	NO	None	
signal_index	int	NO	None	
signal_drift_steps	int	NO	None	
signal_drift_rate	double	NO	None	
signal_snr	double	NO	None	
signal_beam	int	NO	None	
signal_coarse_channel	int	NO	None	
signal_num_timesteps	int	NO	None	
signal_power	double	NO	None	
signal_incoherent_power	double	NO	None	
beam_id	int	NO	MUL	None

In [41]: `%sql SELECT tstart, signal_frequency FROM cosmic_observation_stamp LIMIT 10`

* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations
10 rows affected.

Out[41]:

tstart	signal_frequency
1682436833.3652735	8671.790599822998
1682436833.3652735	8687.789821624756
1682436833.3652735	8699.236042022705
1682436833.3652735	8698.62990951538
1682436833.3652735	8699.236949920654

1682436833.3652735	8500.299907684326
1682436833.3652735	8499.62990951538
1682436833.3652735	8506.234317779541
1682436833.3652735	8507.740589141846
1682436833.3652735	8518.287548065186

Some advanced queries

Now let's look at some stamps with source names starting with 3C*, starting time after June 1, 2023 UTC 00:00:00 and signal frequency between 2000-4000 MHz.

In [42]:

```
%sql SELECT COUNT(file_uri), MAX(tstart), MIN(signal_frequency) \
FROM cosmic_observation_stamp \
WHERE source_name LIKE '3C%' AND tstart > 1685599200 \
LIMIT 100;
```

```
* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations
1 rows affected.
```

Out[42]:

COUNT(file_uri)	MAX(tstart)	MIN(signal_frequency)
177981	1710501846.2064118	4968.827030181885

In [43]:

```
# storing the query results into a database
dat1 = %sql SELECT * \
FROM cosmic_observation_stamp \
WHERE source_name LIKE '3C%' AND tstart > 1685599200 \
AND signal_frequency < 7000.0 \
AND signal_snr < 10.0 \
LIMIT 100;

# Reading the query results into a database
df1 = dat1.DataFrame()
print(df1.head)
```

```
* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations
16 rows affected.
```

```
<bound method NDFrame.head of
and_offset \
0 208670617 21828 AC 512
1 208670670 21828 AC 512
2 208670925 21828 AC 512
3 208670957 21828 AC 512
4 208860918 21828 BD 480
5 208860919 21828 BD 480
6 208860920 21828 BD 480
7 208860943 21828 BD 480
8 208860983 21828 BD 480
9 208860984 21828 BD 480
10 208861007 21828 BD 480
11 208861008 21828 BD 480
...
```

12	208861009	21828	BD	480
13	208861557	21828	BD	480
14	208861558	21828	BD	480
15	208861626	21828	BD	480

	file_uri	file_local_enumeration \
0	/mnt/cosmic-storage-2/data3/batch_processed/14...	23
1	/mnt/cosmic-storage-2/data3/batch_processed/14...	24
2	/mnt/cosmic-storage-2/data3/batch_processed/14...	24
3	/mnt/cosmic-storage-2/data3/batch_processed/14...	24
4	/mnt/cosmic-storage-2/data3/batch_processed/14...	22
5	/mnt/cosmic-storage-2/data3/batch_processed/14...	23
6	/mnt/cosmic-storage-2/data3/batch_processed/14...	24
7	/mnt/cosmic-storage-2/data3/batch_processed/14...	22
8	/mnt/cosmic-storage-2/data3/batch_processed/14...	22
9	/mnt/cosmic-storage-2/data3/batch_processed/14...	23
10	/mnt/cosmic-storage-2/data3/batch_processed/14...	22
11	/mnt/cosmic-storage-2/data3/batch_processed/14...	23
12	/mnt/cosmic-storage-2/data3/batch_processed/14...	24
13	/mnt/cosmic-storage-2/data3/batch_processed/14...	21
14	/mnt/cosmic-storage-2/data3/batch_processed/14...	22
15	/mnt/cosmic-storage-2/data3/batch_processed/14...	22

	source_name	ra_hours	dec_degrees	fch1_mhz	...	signal_index
0	3C286	13.518969	30.509155	5000.325809	...	108305
1	3C286	13.518969	30.509155	5000.240406	...	97111
2	3C286	13.518969	30.509155	5000.240406	...	97111
3	3C286	13.518969	30.509155	5000.240406	...	97111
4	3C286	13.518969	30.509155	5000.235500	...	96468
5	3C286	13.518969	30.509155	5000.321529	...	107744
6	3C286	13.518969	30.509155	5000.486523	...	129370
7	3C286	13.518969	30.509155	5000.235500	...	96468
8	3C286	13.518969	30.509155	5000.235500	...	96468
9	3C286	13.518969	30.509155	5000.321522	...	107743
10	3C286	13.518969	30.509155	5000.321522	...	107743
11	3C286	13.518969	30.509155	5000.295406	...	104320
12	3C286	13.518969	30.509155	5000.486523	...	129370
13	3C286	13.518969	30.509155	5000.429035	...	121835
14	3C286	13.518969	30.509155	5000.235500	...	96468
15	3C286	13.518969	30.509155	5000.235500	...	96468

	signal_drift_steps	signal_drift_rate	signal_snr	signal_beam \
0	0	0.000000	8.069587	2
1	0	0.000000	8.937609	0
2	0	0.000000	8.329020	1
3	0	0.000000	8.199222	4
4	0	0.000000	8.866192	2
5	0	0.000000	8.780730	1
6	0	0.000000	8.239681	4
7	0	0.000000	9.356755	0
8	0	0.000000	9.629790	1
9	1	0.923931	8.149250	3
10	0	0.000000	9.544908	3
11	0	0.000000	9.323606	4
12	0	0.000000	8.270700	2
13	0	0.000000	9.208055	3
14	0	0.000000	8.632877	2
15	0	0.000000	8.565592	2

	signal_coarse_channel	signal_num_timesteps	signal_power \
0	0	64	6.887298e+11
1	0	64	7.052067e+11
2	0	64	6.731105e+11
3	0	64	6.648298e+11
4	31	64	6.664272e+11
5	31	64	6.740623e+11
6	31	64	6.451106e+11
7	31	64	6.972277e+11
8	31	64	7.171040e+11
9	31	64	6.345085e+11
10	31	64	7.034022e+11
11	31	64	6.967816e+11
12	31	64	6.374126e+11
13	31	64	6.875868e+11
14	31	64	6.556076e+11
15	31	64	6.513970e+11

	signal_incoherent_power	beam_id
0	0.0	522617
1	0.0	522621
2	0.0	522658
3	0.0	522670
4	0.0	522635
5	0.0	522634
6	0.0	522637
7	0.0	522627
8	0.0	522622
9	0.0	522624
10	0.0	522618
11	0.0	522619
12	0.0	522617
13	0.0	522669
14	0.0	522668
15	0.0	522659

[16 rows x 34 columns]>

In [44]: `# Let's look at the information of the lowest signal snr row`

```
smin = df1['signal_snr'].argmin()
smax = df1['signal_snr'].argmax()
print(smin, df1['signal_snr'][smin], smax, df1['signal_snr'][smax])
```

```
0 8.069586753845215 8 9.629790306091309
```

In [45]:

```
# let's get the stamp of the lowest snr signal
df1.iloc[smin,:].head
file = df1.iloc[smin,:]['file_uri']
num = df1.iloc[smin,:]['file_local_enumeration']
print(f" Enumeration number:{num}, Stamp file: {file}")
```

```
Enumeration number:23, Stamp file: /mnt/cosmic-storage-2/data3/batch_processed/14A-000.sb45284449.eb45302097.60384.45817337963.15.1/14A-000.sb45284449.eb45302097.60384.45817337963.15.1.AC.C512.0000.raw.seticore.0000.stamps
```

Use Nested Queries, UNION and JOIN

UNION is used combine search results from mutiple tables whereas JOIN is used combine rows from multiple tables based on conditions.

Let's look at some observation datasets which has poor calibration grades (less than 0.4 grade)

In [46]:

```
# For this I think we need to do a nested query

%sql SELECT cosmic_observation.scan_id \
FROM cosmic_observation \
WHERE cosmic_observation.id = (SELECT observation_id \
FROM cosmic_observation_calibration \
WHERE overall_grade < 0.4 \
LIMIT 1) ;

# we can use IN to get a list of scan Ids. However this version of MySQL
```

```
* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations
1 rows affected.
```

Out[46]:

scan_id
VLASS3.1.sb43866247.eb44049643.60098.68515746528.199.1

In [47]:

```
# Now combining the first 5 source names from cosmic_observation_beam t

#UNION ALL combines duplicate results

%sql SELECT source\
FROM cosmic_observation_beam \
WHERE id < 5 \
UNION ALL \
SELECT source_name\
FROM cosmic_observation_hit \
```

```
WHERE id < 10;
```

```
* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations  
13 rows affected.
```

Out[47]:

source

PHASE_CENTER

PHASE_CENTER

PHASE_CENTER

PHASE_CENTER

PHASE_CENTER

PHASE_CENTER

PHASE_CENTER

Incoherent

Incoherent

Incoherent

Incoherent

PHASE_CENTER

PHASE_CENTER

In [48]:

```
# UNION combines only distinct results  
# combining source names from the cosmic_observation_beam table and cos
```

```
%sql SELECT source\  
FROM cosmic_observation_beam \  
WHERE id < 50 AND source <> "Incoherent"\  
UNION \  
SELECT source_name\  
FROM cosmic_observation_hit \  
WHERE id < 50 AND source_name <> "Incoherent";
```

```
* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations  
13 rows affected.
```

Out[48]:

source

PHASE_CENTER

2563750130277249920

2563750989270708352

2563705565696622208

2563755456036730880

2563753011208212011

2563702335881216128

2563708039597806080

2539927664673403520

2539934880218459392

2539937865220098304

2539934257447574784

2539919246537470848

In [49]: *JOIN to join rows from multiple tables based on conditions*
Joining results from cosmic_observation_hit table and cosmic_observation
`sql` SELECT cosmic_observation_hit.signal_frequency, cosmic_observation_hi
FROM cosmic_observation_hit \
JOIN cosmic_observation \
ON cosmic_observation_hit.observation_id = cosmic_observation.id \
LIMIT 10;

* mysql+pymysql://cosmic:***@cosmic-storage-1/cosmic_observations
10 rows affected.

Out[49]:

signal_frequency	signal_snr	
15031.645931243896	36.756961822509766	000.sb43813989.eb43862811.60069.2774
15031.753147125244	8.49901294708252	000.sb43813989.eb43862811.60069.2774
15031.84638595581	8.693612098693848	000.sb43813989.eb43862811.60069.2774
15031.645931243896	38.80583572387695	000.sb43813989.eb43862811.60069.2774
15031.753147125244	9.229655265808105	000.sb43813989.eb43862811.60069.2774
15050.263317108154	8.006394386291504	000.sb43813989.eb43862811.60069.2774
15050.2766456604	20.34689712524414	000.sb43813989.eb43862811.60069.2774
15050.263317108154	8.493431091308594	000.sb43813989.eb43862811.60069.2774
15050.2766456604	22.371721267700195	000.sb43813989.eb43862811.60069.2774